

C Programming: 100 Questions to Master C (From Basics to Advanced)

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Section 1: Basics of C (Q1Q10)

1. Write a C program to print "Hello, World!".
2. Write a C program to add, subtract, multiply and divide two numbers.
3. Write a program to find the ASCII value of a character.
4. Write a program to swap two numbers using a temporary variable.
5. Write a program to find the size of different data types using sizeof().
6. Write a program to convert temperature from Celsius to Fahrenheit.
7. Write a program to calculate the area and perimeter of a rectangle.
8. Write a program to use all arithmetic and relational operators.
9. Write a program to demonstrate increment and decrement operators.
10. Write a program to take input from user and display it.

Section 2: Control Flow (Q11Q20)

11. Write a program to check if a number is even or odd.
12. Write a program to find the maximum of three numbers using if.
13. Write a program to check if a character is a vowel or consonant.
14. Write a program to find whether a year is a leap year or not.
15. Write a program to implement a basic calculator using switch.
16. Write a program to print the multiplication table of a number.
17. Write a program to print all prime numbers between 1 and 100.
18. Write a program to find factorial of a number using for loop.
19. Write a program to reverse a number using while loop.
20. Write a program to find the sum of digits of a number.

Section 3: Functions (Q21Q30)

21. Write a function to calculate the square of a number.
22. Write a function to check if a number is prime.
23. Write a function to find the GCD of two numbers.
24. Write a recursive function to calculate factorial of a number.

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25. Write a program with a function to reverse a string.
26. Write a program to demonstrate pass by value and pass by reference.
27. Write a program to calculate nCr using functions.
28. Write a program to count the number of digits using recursion.
29. Write a program to use function with default-like behavior using macros.
30. Write a function to check whether a number is an Armstrong number.

Section 4: Arrays & Strings (Q31Q45)

31. Write a program to read and print elements of an array.
32. Write a program to find the sum and average of array elements.
33. Write a program to find the largest and smallest elements in an array.
34. Write a program to reverse an array in-place.
35. Write a program to sort array elements in ascending order.
36. Write a program to perform matrix addition.
37. Write a program to perform matrix multiplication.
38. Write a program to check if a matrix is symmetric.
39. Write a program to count vowels and consonants in a string.
40. Write a program to reverse a string using array.
41. Write a program to check whether two strings are equal.
42. Write a program to convert lowercase string to uppercase.
43. Write your own implementation of `strlen()`.
44. Write a program to remove duplicate characters from a string.
45. Write a program to count frequency of each character in a string.

Section 5: Pointers (Q46Q60)

46. Write a program to demonstrate pointer basics (address and value).
47. Write a program to swap two variables using pointers.
48. Write a program to find the length of a string using pointers.
49. Write a program to copy one string to another using pointers.
50. Write a program to reverse a string using pointers.
51. Write a program to access array elements using pointers.
52. Write a program to find sum of array elements using pointers.

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- 53. Write a program to demonstrate pointer to pointer.
- 54. Write a program to find the max of two numbers using pointer.
- 55. Write a program to allocate memory dynamically using malloc().
- 56. Write a program to allocate memory using calloc() and free it.
- 57. Write a program to resize memory block using realloc().
- 58. Write a program to compare pointer arithmetic (int vs char).
- 59. Write a program to return pointer from a function.
- 60. Write a program to demonstrate memory leak and fix it.

Section 6: Structures & Unions (Q61Q75)

- 61. Write a program to define and use a structure for student data.
- 62. Write a program to pass structure to a function.
- 63. Write a program to return structure from a function.
- 64. Write a program to use array of structures.
- 65. Write a program to use nested structures.
- 66. Write a program to sort an array of structures by age or marks.
- 67. Write a program to use pointers to structures.
- 68. Write a program to demonstrate structure with typedef.
- 69. Write a program to demonstrate union and compare it with structure.
- 70. Write a program using enum to represent days of the week.
- 71. Write a program to use bit-fields in structures.
- 72. Write a program to demonstrate size of structure with padding.
- 73. Write a program to compare memory usage of struct vs union.
- 74. Write a program using a structure to implement complex number operations.
- 75. Write a program to store employee records and search by ID.

Section 7: File Handling (Q76Q85)

- 76. Write a program to create a text file and write data to it.
- 77. Write a program to read contents from a file.
- 78. Write a program to count characters, words, and lines in a file.
- 79. Write a program to copy one file to another.
- 80. Write a program to merge two text files into a third file.

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81. Write a program to update contents in a file.
82. Write a program to delete specific content from a file.
83. Write a program to read a file and print only numbers.
84. Write a program to encrypt and decrypt text using files.
85. Write a program to count frequency of each word in a file.

Section 8: Advanced Concepts (Q86Q100)

86. Write a program to define and use macros with arguments.
87. Write a program to demonstrate conditional compilation.
88. Write a program to pass command-line arguments and print them.
89. Write a program to sort command-line arguments alphabetically.
90. Write a program to implement a simple shell using `system()`.
91. Write a program to demonstrate bitwise AND, OR, XOR, NOT.
92. Write a program to set, clear, toggle specific bit in a number.
93. Write a program to count the number of set bits.
94. Write a program to implement function pointers.
95. Write a program to sort an array using function pointer to comparator.
96. Write a program to demonstrate modular programming with multiple files.
97. Write a program using assertions for input validation.
98. Write a program to implement a singly linked list in C.
99. Write a program to reverse a linked list.
100. Write a program to implement stack using arrays and structures.